DO NOW — RECALL

Lesson 068 · Backtracking: Choices, Dead Ends and Und

Name: _____ Class: _____ Date: _____

1. Describe the divide-and-conquer approach to solving a problem.

2. What is a dead end in a search?

3. Why must recursion have a base case?

4. Stack Push 2, Push 5, Pop, then Push 8: give popped item and remaining stack.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Divide the problem into smaller subproblems, solve them, then combine their solutions to solve the original problem.

2. Model answer

A state unable to reach any valid solution.

3. Model answer

To stop further calls at a terminal state.

4. Model answer

Popped 5; bottom-to-top [2,8].

By the end, you can…

• Backtracking explores a choice, tests whether its state remains viable, then undoes the choice if it cannot lead to a solution.

• A search tree records choices and alternative branches. Pruning avoids expanding branches already proven invalid.

• A correct search needs a stopping condition. Pruning on sum>target only works when remaining numbers cannot reduce the sum (e.g. non-negative inputs).

Vocabulary — know these terms

backtracking — Try and undo choices.

search tree — Branches of possible decisions.

pruning — Stop provably impossible branch.

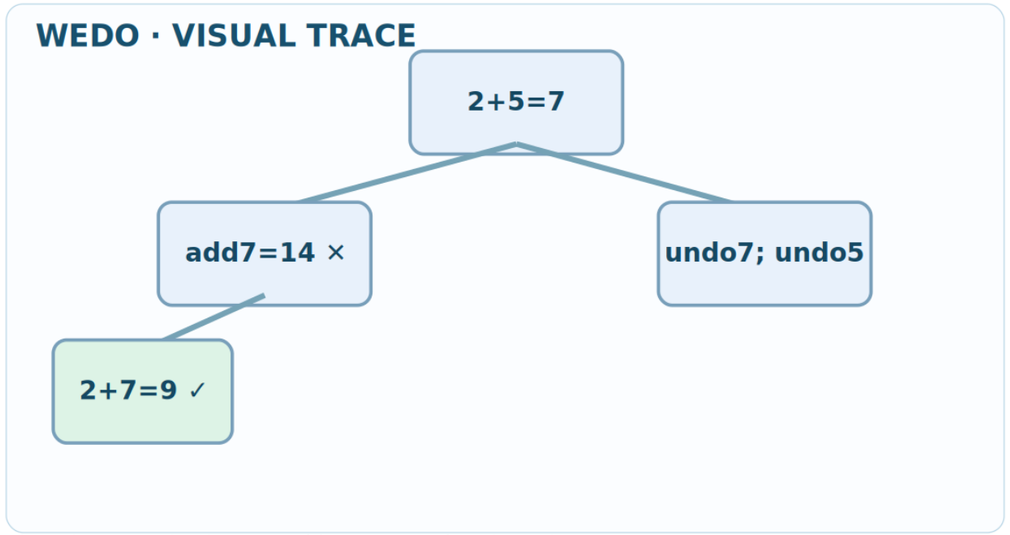
state — Current partial solution data.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find a subset of [2,5,7] with target 9.

The alternative branch 2+7 gives a valid solution.



WEDO · VISUAL TRACE; 2+5=7; add7=14 ×; undo5; 2+7=9 ✓

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Follow the reasoning
The alternative branch 2+7 gives a valid solution.

NEW LEARNING

Read, annotate and explain

Define and predict

Backtracking explores a choice, tests whether its state remains viable, then undoes the choice if it cannot lead to a solution.

Explain

A search tree records choices and alternative branches. Pruning avoids expanding branches already proven invalid.

Evaluate

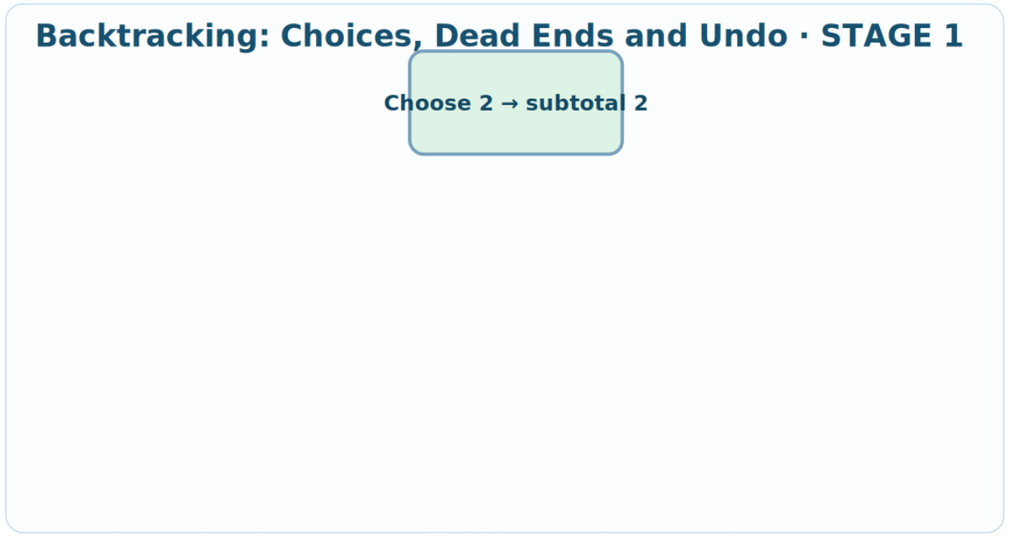
A correct search needs a stopping condition. Pruning on sum>target only works when remaining numbers cannot reduce the sum (e.g. non-negative inputs).

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Backtracking: Choices, Dead Ends and Undo · STAGE 1; Choose 2 → subtotal 2

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A branch is always impossible when its subtotal exceeds the target, even if negatives remain.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

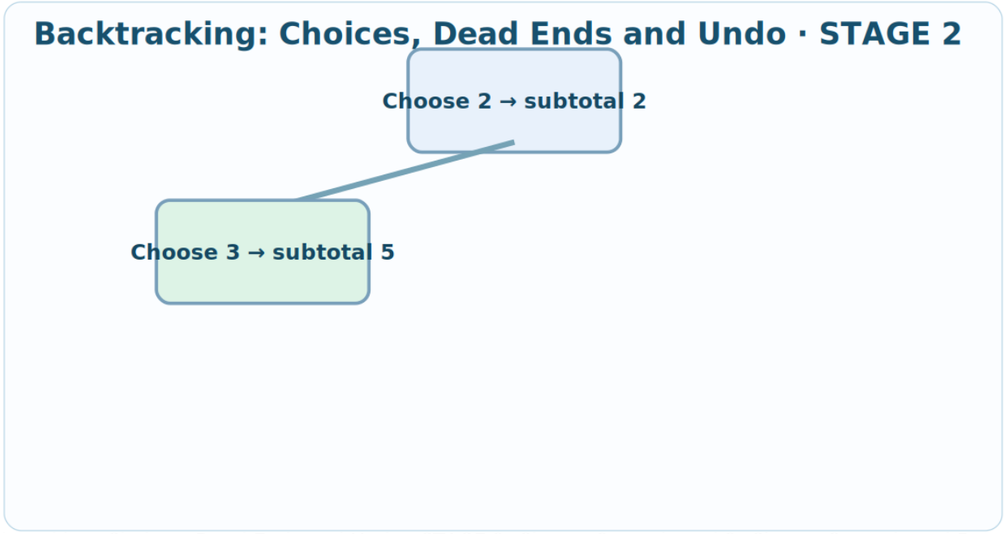
Question 3

Explain why pruning when sum>target may be invalid if negative integers occur; give counterexample. [6]

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

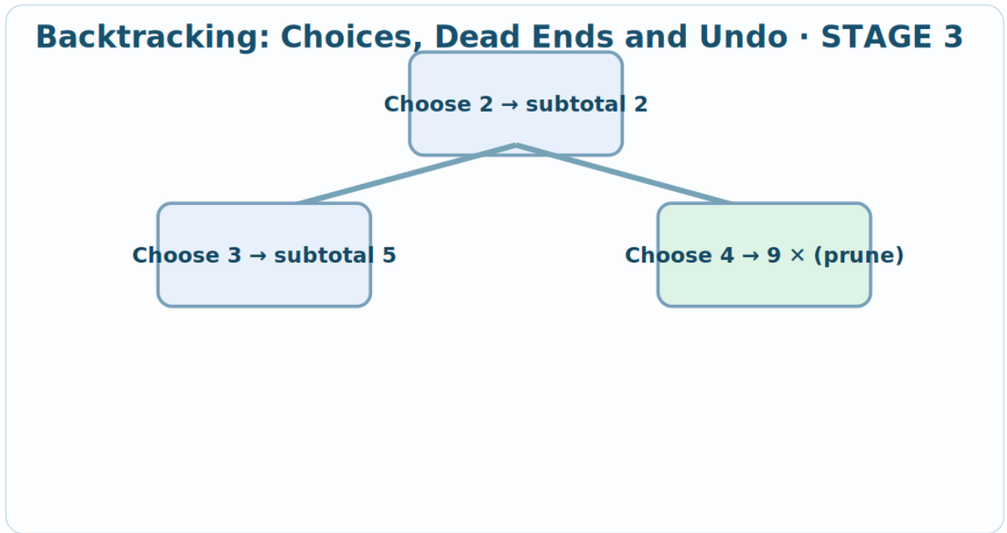


Backtracking: Choices, Dead Ends and Undo · STAGE 2; Choose 2 → subtotal 2; Choose 3 → subtotal 5

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Backtracking: Choices, Dead Ends and Undo · STAGE 3; Choose 2 → subtotal 2; Choose 3 → subtotal 5; Choose 4 → 9 × (prune)

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Trace first failed and one successful branch for subset sum target6 from [2,3,4]. [6]

Question 2

Explain a recursive backtracking design with choose/exclude and two stopping conditions. [6]

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

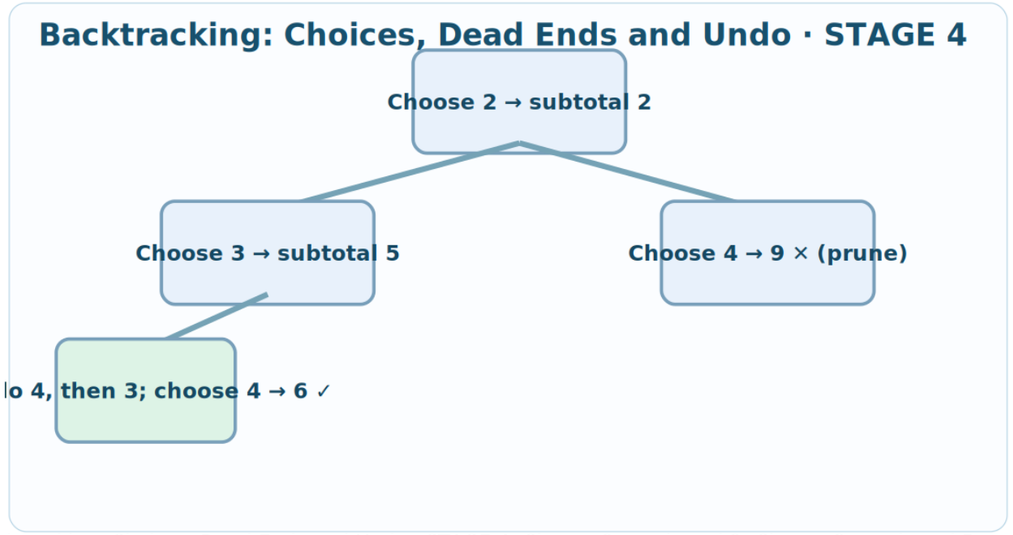
Guided 1 — TRY

Find a subset of [2,5,7] with target 9.
Source code / input:
Choose2,5 →7; add7 →14 reject. Undo7 →7; exclude7 fails at end. Undo5 →2; choose7 →9.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Backtracking: Choices, Dead Ends and Undo · STAGE 4; Choose 2 → subtotal 2; Choose 3 → subtotal 5; Choose 4 → 9 × (prune); Undo 4, then 3; choose 4 → 6 ✓

I DO — WORKED EXAMPLE

Follow the worked method

Find a subset of [1,3,4] with total 5.

Follow the reasoning

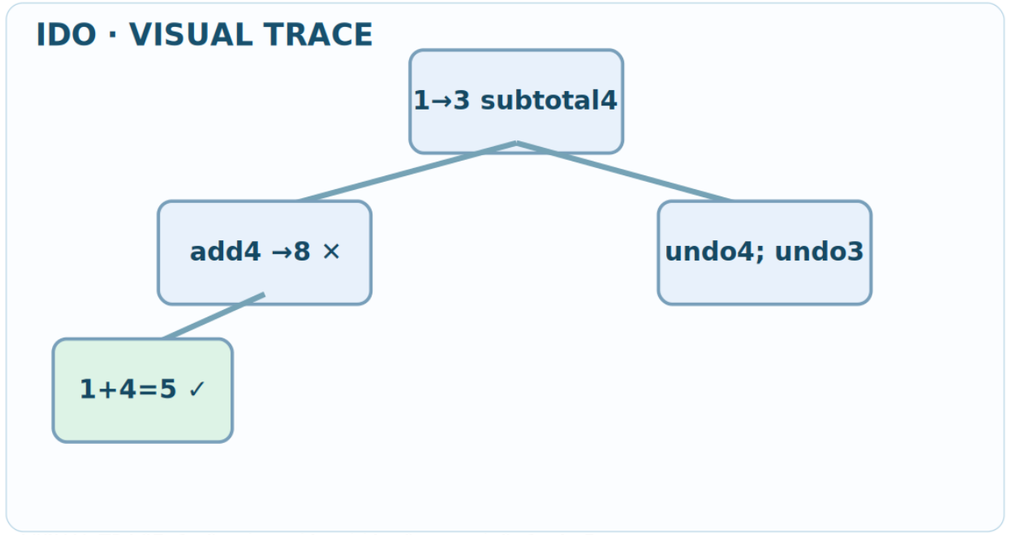
Return from the failed choice of4 before excluding3. The state returns to [1], then choosing4 reaches5. Recursive local totals restore on return; mutable state needs an explicit undo.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find a subset of [1,3,4] with total 5.

Return from the failed choice of4 before excluding3. The state returns to [1], then choosing4 reaches5. Recursive local totals restore on return; mutable state needs an explicit undo.



IDO · VISUAL TRACE; 1→3 subtotal4; add4 →8 ×; undo3; 1+4=5 ✓